🔑 **Answer Key — Teacher Copy · Lesson 3-6**

MSTAR PRACTICE WORKSHEET

Lesson 3-6: Ratio Reasoning: Convert Measurements within the Same System

Grade 6 Mathematics · Standard 6.AT.3

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: D

There are 12 inches in 1 foot. How many inches are in 5 feet?

A. 7 inches

B. 17 inches

C. 50 inches

D. 60 inches

Inches are smaller than feet, so multiply by the unit ratio: 5 × 12 = 60 inches.

Question 2 SELECTED RESPONSE — CORRECT: D

There are 1,000 milliliters in 1 liter. How many milliliters are in 3 liters?

A. 30 mL

B. 300 mL

C. 1,003 mL

D. 3,000 mL

Multiply by the unit ratio: 3 × 1,000 = 3,000 milliliters.

Question 3 SELECTED RESPONSE — CORRECT: A

There are 16 ounces in 1 pound. How many pounds is 48 ounces?

- (A) 3 pounds
- (B) 32 pounds
- (C) 64 pounds
- (D) 768 pounds

Pounds are the LARGER unit, so divide: 48 ÷ 16 = 3 pounds.

Question 4 SELECTED RESPONSE — CORRECT: C

A child is 4 feet tall. A ride requires 44 inches. Can the child ride?

- (A) No, because 4 is less than 44
- (B) Yes, because 4 feet = 44 inches exactly
- (C) Yes, because 4 feet = 48 inches, which is more than 44
- (D) You cannot compare feet and inches

Convert first so both measurements use the same unit: 4 × 12 = 48 inches. 48 > 44, so the child can ride.

Question 5 SELECTED RESPONSE — CORRECT: C

Diamond needs 4 liters of water. Her bottle holds 500 milliliters. How many times must she fill it?

- (A) 2 times
- (B) 4 times
- (C) 8 times
- (D) 40 times

Convert first: 4 L = 4,000 mL. Then solve ? × 500 = 4,000, giving 8 fills.

Question 6 SELECTED RESPONSE — CORRECT: D

How many inches are in 2 feet?

A 10

B 12

C 14

D 24

2 × 12 = 24 inches.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: D

A rope measures 7 feet. Using 1 foot = 12 inches, how long is the rope in inches?

A 0.58 inches

B 7 inches

C 19 inches

D 84 inches

7 feet × 12 inches per foot = 84 inches.

- **A:** That divides 7 by 12, converting the wrong direction. Inches are smaller than feet, so the count gets bigger.
- **B:** That leaves the measurement in feet. Each foot holds 12 inches, so multiply.
- **C:** That adds 7 and 12. Converting to a smaller unit multiplies: 7 x 12.

Part B — correct answer: C

How do you know the answer to Part A should be a LARGER number than 7?

A Converting always makes numbers larger.

B Multiplying always makes numbers larger.

C An inch is smaller than a foot, so it takes more inches to measure the same length.

D Feet and inches are in different systems.

The smaller the unit, the more of them are needed to cover the same length. Since an inch is smaller than a foot, the count in inches must be larger.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Converting the Wrong Direction

> Tariq converted 240 centimeters to meters. Knowing 1 meter = 100 centimeters, he wrote: '240 × 100 = 24,000 meters.'

Explain Tariq's misconception and give the correct conversion.

Model answer: 240 ÷ 100 = 2.4 meters. A meter is larger than a centimeter, so fewer of them cover the same length — the number must get smaller, which means dividing, not multiplying. 24,000 meters is 24 kilometers.

2 points	Student identifies that a meter is LARGER than a centimeter so the number of meters must be smaller, computes 240 ÷ 100 = 2.4 meters, and explains why division is correct here.
1 point	Student gives 2.4 meters without explaining the size relationship.
0 points	Student agrees with Tariq or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

Michael's mother is setting four fence posts. Each post needs 250 ounces of concrete mix, one bag holds 80 ounces, and each bag needs 1 quart of water. How many bags and how many cups of water does she need? Show your conversions.

Model answer: Total mix: 4 posts × 250 ounces = 1,000 ounces. Bags needed: 1,000 ÷ 80 = 12.5, and you cannot buy half a bag, so she needs 13 bags. Water: each bag takes 1 quart, so 13 bags take 13 quarts. Converting to cups with the unit ratio 4 cups : 1 quart, 13 × 4 = 52 cups of water. The two conversions go opposite ways: ounces to bags divides (a bag is the larger unit), while quarts to cups multiplies (a cup is the smaller unit).

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.